



NEXUS EARTH

AI-Powered Digital Twin & Decision Intelligence

Building Tomorrow Before It Happens.

Simulate

Understand

Decide

nexus-earth.netlify.app/command



The Global Problem

Global crises are connected — one shock does not stay contained.

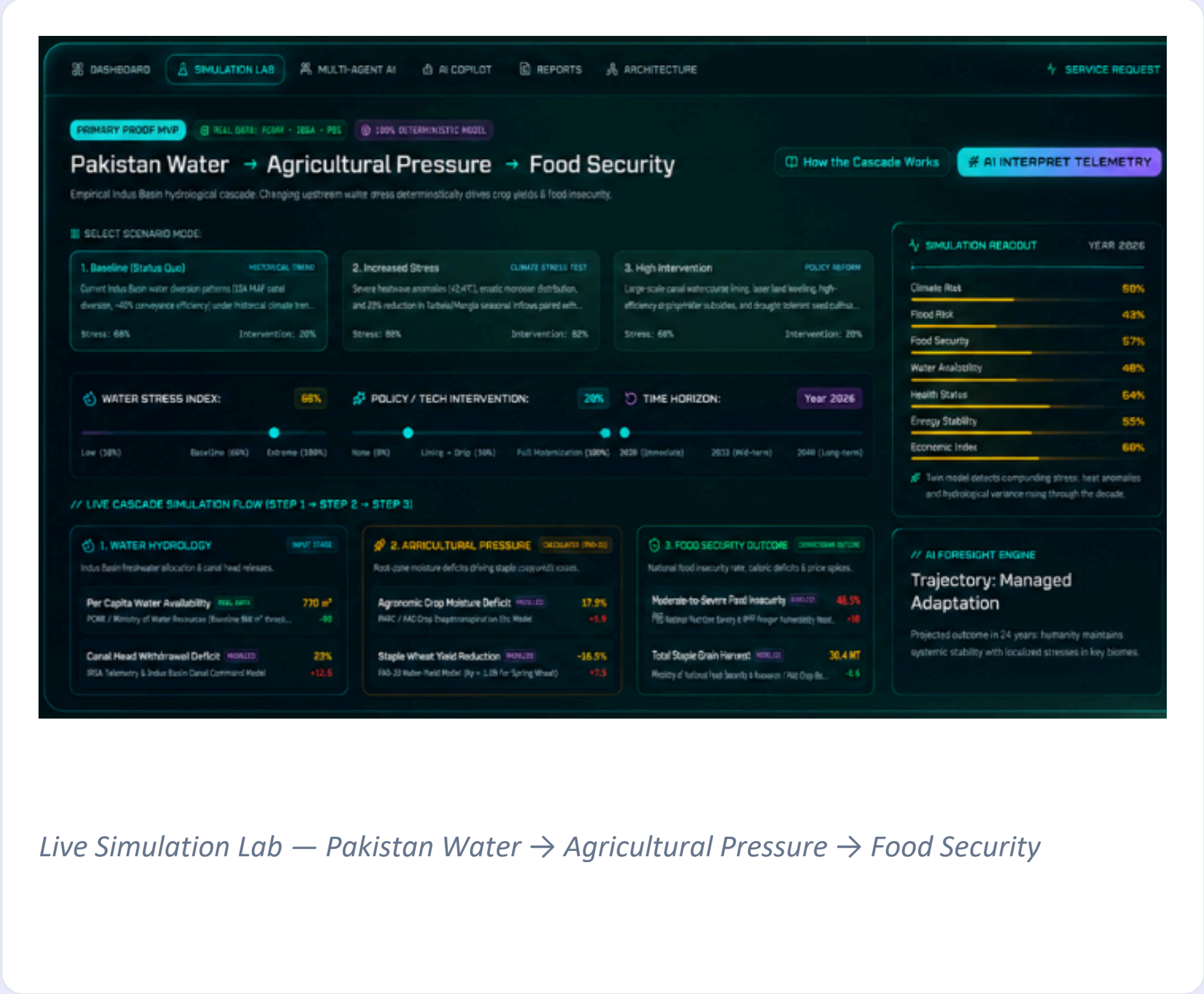
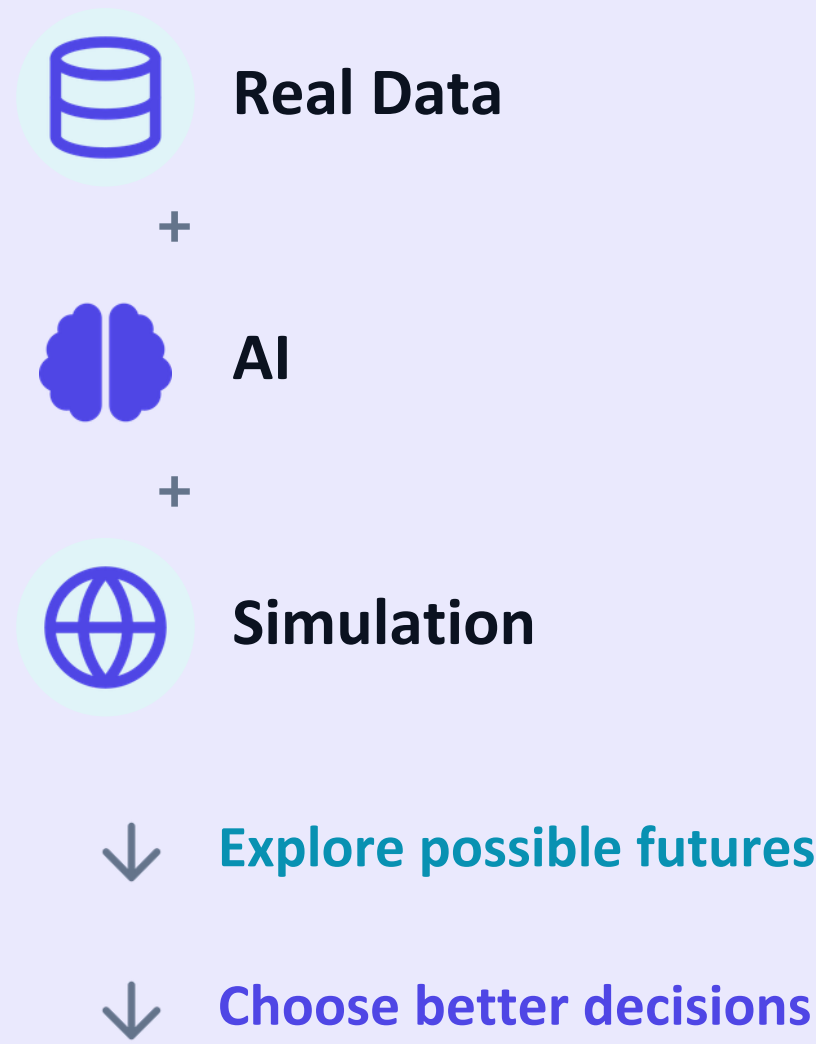


One crisis can trigger the next.

Floods • Water Scarcity • Food Insecurity • Energy Pressure • Climate Disasters

THE IDEA

What If We Could Test a Decision Before Reality Does?



Live Simulation Lab — Pakistan Water → Agricultural Pressure → Food Security

Who Is Nexus Earth For?

PRIMARY USERS



Government & Policy Makers



Water & Food Authorities



Disaster & Climate Response Teams

SECONDARY USERS



Researchers



Urban & Strategic Planners



Built for decision-makers whose choices affect millions.

What Is Nexus Earth?

A Digital Twin of Humanity



Climate



Water



Food



Energy

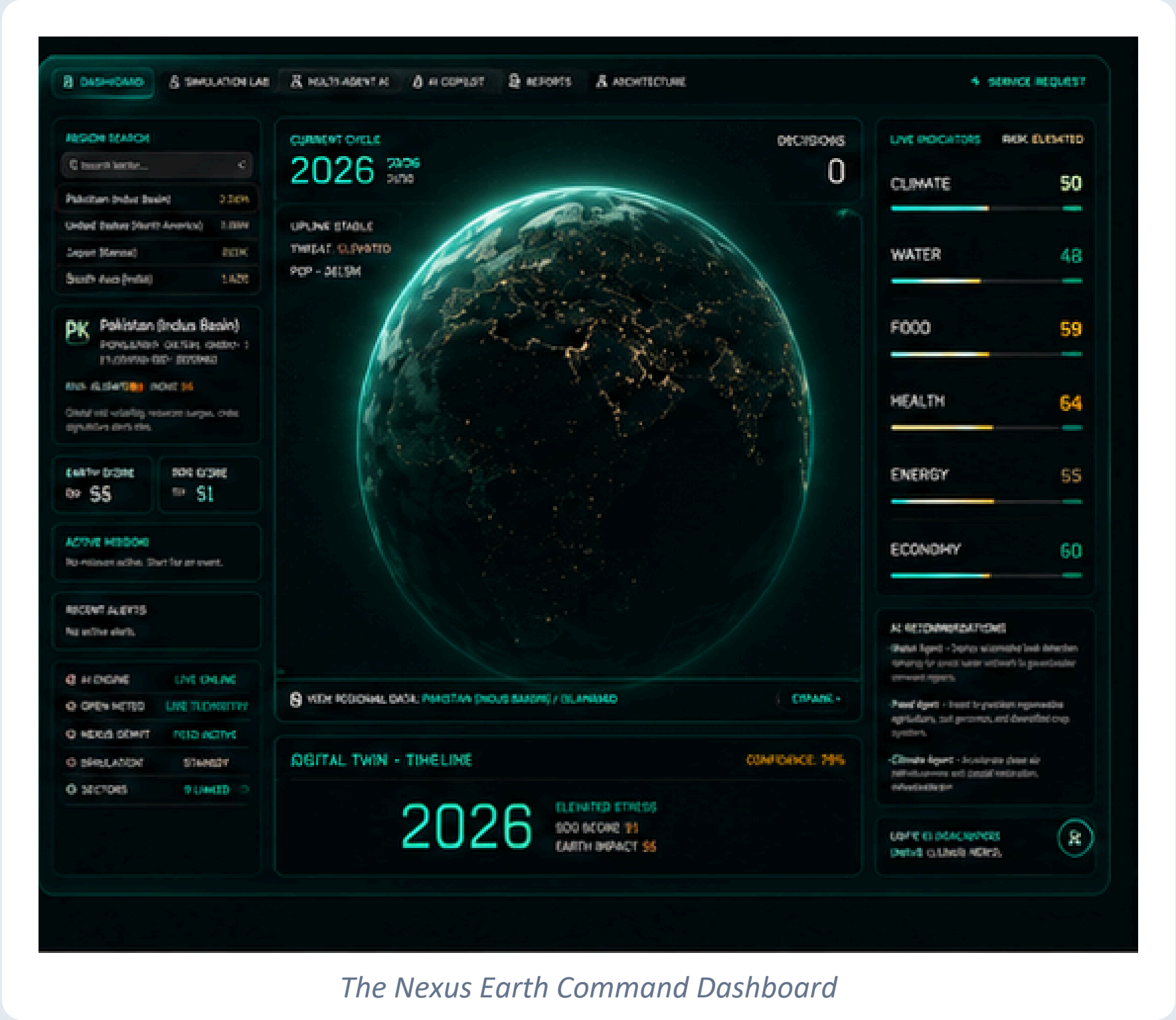


Health



Economy

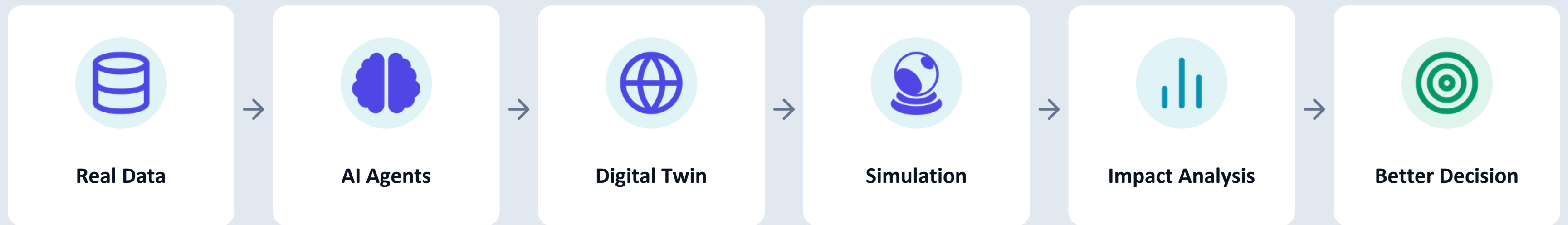
 One **interactive** decision system.



The Nexus Earth Command Dashboard

How It Works

From Data to Decision



Choose a problem → Test an intervention → Simulate consequences → Make a better decision

This is the actual user workflow inside Nexus Earth's Simulation Lab.

CAPABILITIES

What Can It Do?



AI Copilot

Ask → Analyze → Recommend



Simulation

Change decisions → Explore outcomes



Cause & Effect

See how crises connect



Earth Memory

Learn from past events

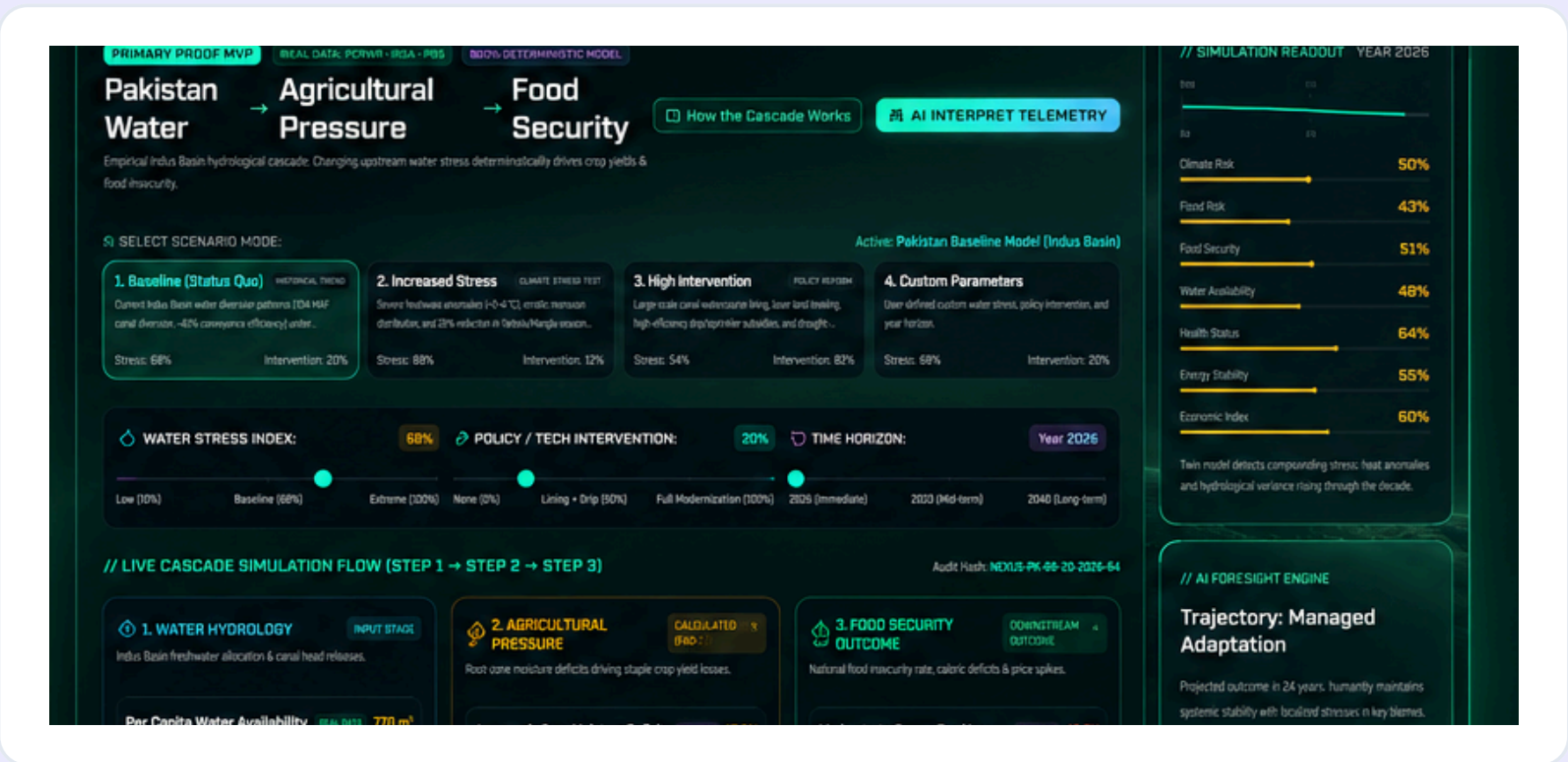
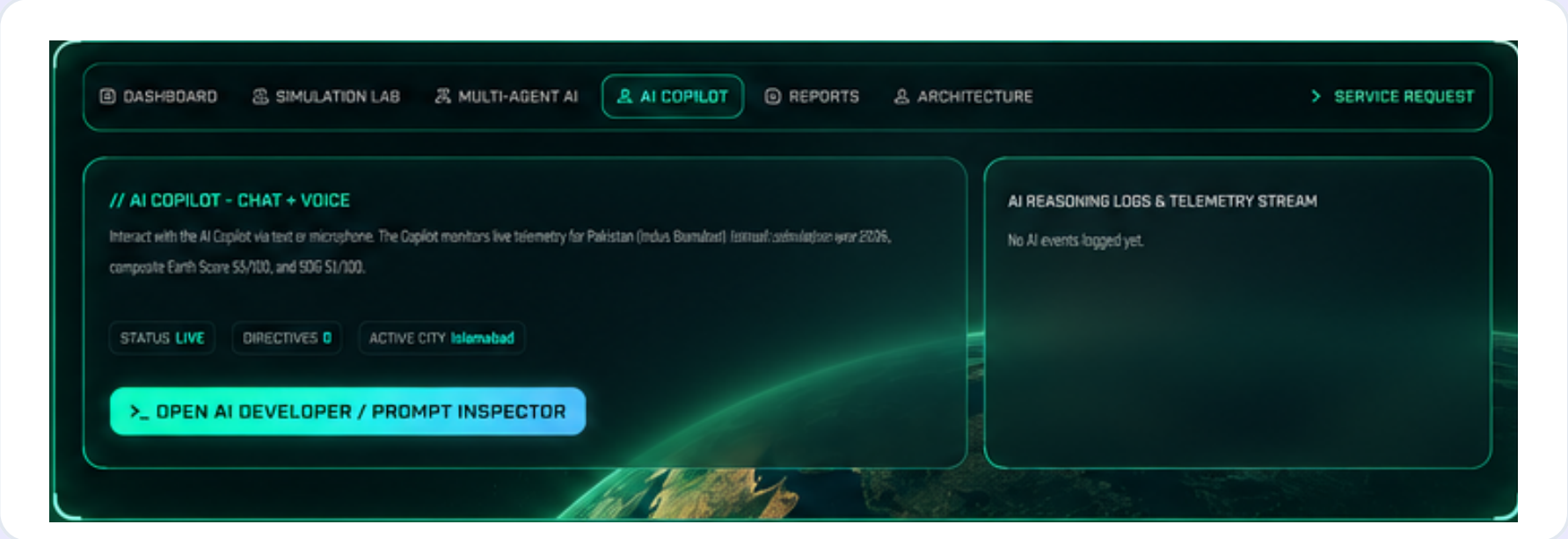


Service Request

Report a problem → Connect to crisis system

Less Paperwork • Less Time Wasted • Less Money Wasted

One decision.
Multiple consequences.



UNDER THE HOOD

Multi-Agent AI & Architecture

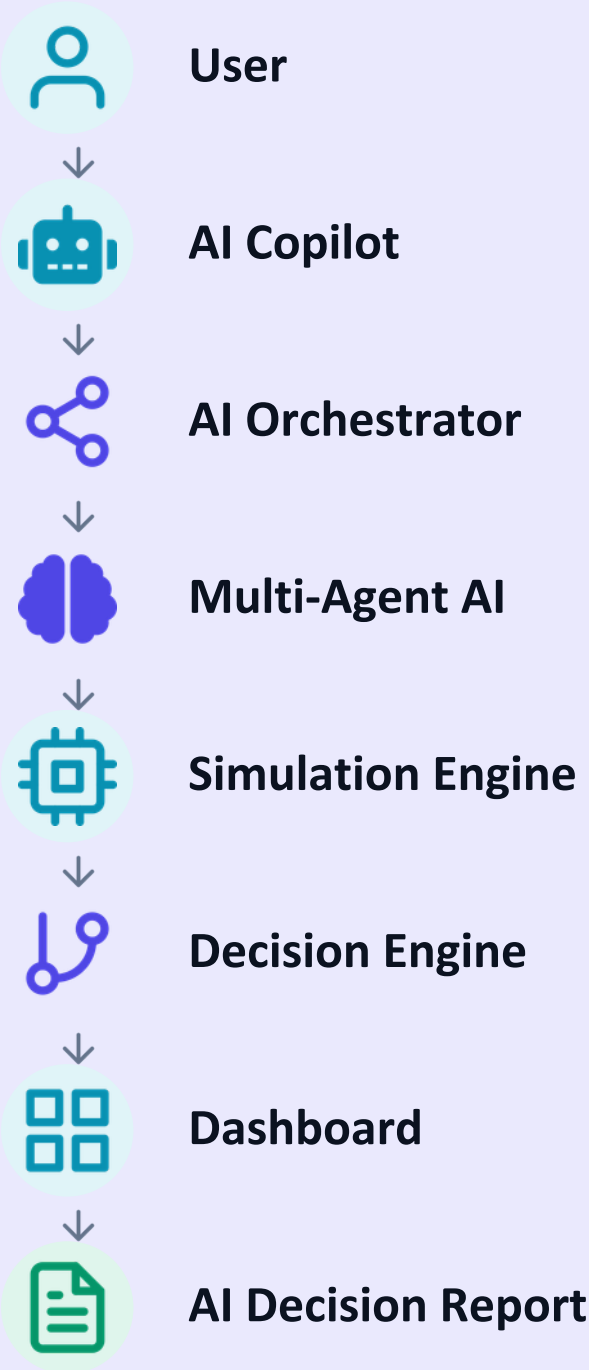
SIX SPECIALIZED AGENTS

Each agent monitors one domain and feeds insight to the orchestrator.

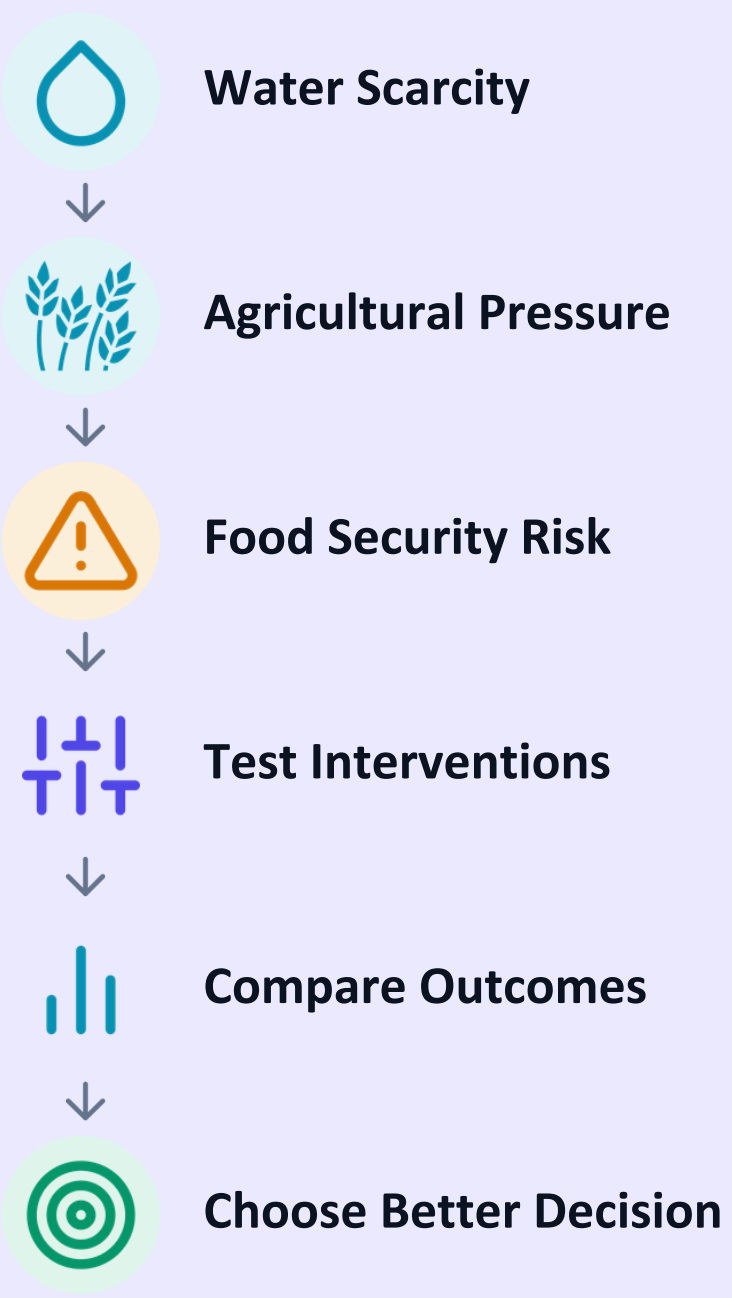


Multi-Agent AI = 6 domain experts reasoning together, not one generic model.

FROM USER TO DECISION



Pakistan Water Crisis



DASHBOARD

SIMULATION LAB

MULTI-AGENT AI

AI COPILOT

REPORTS

ARCHITECTURE

SERVICE REQUEST

PRIMARY PROOF MVP

REAL DATA: POTWRI - IRSA - PBS

800% DETERMINISTIC MODEL

Pakistan Water

Agricultural Pressure

Food Security

How the Cascade Works

AI INTERPRET TELEMTRY

Empirical Indus Basin hydrological cascade. Changing upstream water stress deterministically drives crop yields & food insecurity.

SELECT SCENARIO MODE:

Active: Pakistan Baseline Model (Indus Basin)

1. Baseline (Status Quo)

2. Increased Stress

3. High Intervention

4. Custom Parameters

WATER STRESS INDEX: 68%

POLICY / TECH INTERVENTION: 20%

TIME HORIZON: Year 2026

1. WATER HYDROLOGY

2. AGRICULTURAL PRESSURE

3. FOOD SECURITY OUTCOME

Per Capita Water Availability

Agronomic Crop Moisture Deficit

Moderate-to-Severe Food Insec...

Simulation Readout Year 2026

Climate Risk: 50%

Flood Risk: 43%

Food Security: 51%

Water Availability: 48%

Health Status: 64%

Energy Stability: 55%

Economic Index: 60%

AI Foresight Engine

Trajectory: Managed Adaptation

Live Simulation Lab: Pakistan Indus Basin — modeling how water stress cascades into agricultural and food-security outcomes.

The Impact



Predict Earlier

Identify cascading risks before they escalate.



Test Decisions

Compare possible interventions in simulation.



Act Smarter

Choose strategies based on projected outcomes.

From reactive decisions → to evidence-driven decisions.



CLOSING

Don't just predict the future.
Simulate it.

NEXUS EARTH

Building Tomorrow Before It Happens.

Areesha Iftikhar • Maryam Saleem • Taqi Agha



Live Demo: nexus-earth.netlify.app/command

